

Synthetic Channel Assessment

Compare redundant observations without overstating evidence

Eight fixed learner-operated channels; no external load

Lesson	028 — component	Time	180–240 minutes
Prerequisites	Lessons 001, 002, 007	Board	Arduino Mega 2560
Evidence	startup sweep, three staged wins, state cadences	Status	host verified
Energy class	E1: USB, passive buttons, resistor-limited LEDs		

Boundary. Every observation is synthetic and selected by the learner. “Short-simulated” is a switch position, not an electrical diagnosis. This circuit contains no relay, launcher, initiator, external load, radio, captured protocol, or generic actuation sink.

1 Mission and model

The lesson preserves a short, inspectable chain:

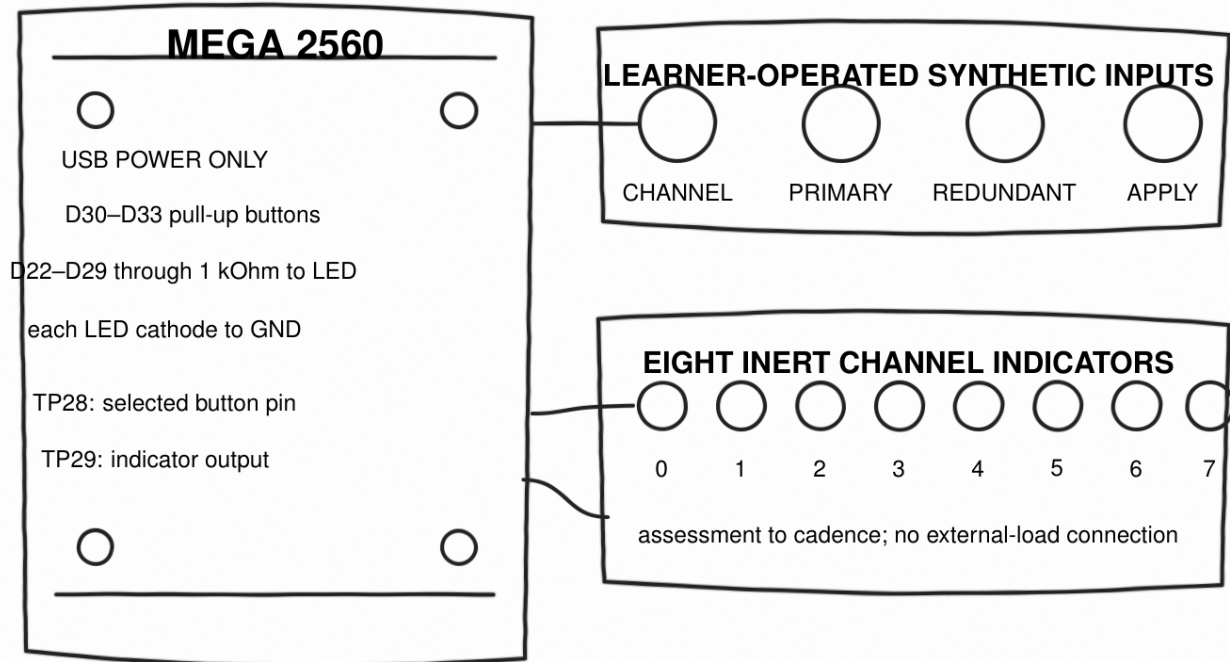
primary + redundant + local age → assessment → LED cadence.

An assessment records what the supplied model says. It does not establish physical continuity, safety, provenance, or permission to energize anything. The core returns values; only the example maps those values to low-voltage indicators.

1.1 Learning outcomes

- derive a state from two explicit observations and local age;
- explain why unavailable, contradictory, and stale are distinct;
- submit a complete set transactionally;
- replay immutable timestamped sets without a catch-up burst;
- separate resource-acquisition evidence from safe-state evidence;
- state precisely what this simulation cannot prove.

2 Orientation and unpowered inspection



Orientation plate: verify exact pins unpowered against the canonical sketch.

Alternative text: graphite orientation plate showing one USB-powered Mega, four pull-up buttons on D30–D33, eight separately resistor-limited LEDs on D22–D29, input test point TP28, and indicator test point TP29. No external-load interface is present.

Role	Mega pin	Connection
channel indicators	D22–D29	pin to 1 k Ω , LED anode; cathode to GND
channel, primary, redundant, apply	D30–D33	momentary switch to GND; internal pull-up
TP28	selected input pin	meter or logic probe relative to Mega GND
TP29	selected indicator pin	meter or logic probe relative to Mega GND

Unpowered check. Confirm one resistor per LED, LED polarity, switch-to-ground wiring, and common ground. Nothing leaves the USB-powered low-voltage teaching circuit.

3 Truth table before code

Primary	Redundant	Assessment	Why
open	open	open	same usable value
closed	closed	closed	same usable value
short-simulated	short-simulated	short-simulated	same synthetic selection
two different usable values	—	contradictory	evidence disagrees
either unavailable	any	unavailable	one required observation is absent
either older than threshold	any	stale	age dominates classification

Age equal to **stale** After retains the underlying state. The first greater age is stale. The half-range timestamp rule prevents an ambiguous future value from being mistaken for old evidence.

4 Lifecycle and transactional update

Construction is inert. `initialize()` rejects a zero or ambiguous stale interval, repeated initialization is harmless, `shutdown()` is idempotent, and destruction calls shutdown. No heap, exception, RTTI, hidden clock, callback, or hardware owner appears in the assessor.

`update()` first validates the complete submission. It rejects:

- more than eight entries or null storage with a nonzero count;
- an ID outside 0–7 or the same ID twice;
- an invalid observation enum;
- a timestamp in the ambiguous future half of the 32-bit space.

Only after all entries pass does it replace slots. Therefore a bad final entry cannot partly overwrite earlier channels.

For example, if a submitted set ends with a duplicate channel, the entire set is rejected and every preceding channel keeps its earlier assessment. An omitted channel likewise retains its last accepted observation.

5 Recorded evidence source

`RecordedInertObservationSource` walks caller-owned immutable sets. Each set holds all eight observations and one due time. Times are strictly increasing and remain within the unambiguous 32-bit interval.

```
source.update    (now);
snapshot = source.snapshot ();
assessor.update (now, snapshot.observations,
                  snapshot.observationCount);
```

A delayed update selects the latest due complete set. It never emits each missed set as a burst. Repeated snapshots expose the same pointers and values until another set becomes due. The caller must keep the immutable storage alive longer than the source.

For sets due at 100, 200, and 300 ms, calls at 90, 250, 250, and 400 ms expose no set, the 200 ms set, the same 200 ms set, and the 300 ms set. The delayed 250 ms call coalesces rather than replaying the missed 100 ms set.

6 Narrative Mega example

```
observeOperator   (now);
decideAssessments (now);
showAssessments   (now);
```

Objects appear in dependency order: runtime, buttons, indicators, assessor, then observation values. Setup acquires every owner, configures all channels open, and starts the panel. The loop observes debounced events, decides only when Apply submits a complete set, then presents all assessments.

At startup, one light travels across all eight indicators for two seconds. That response proves that acquisition reached the presentation stage; it is not a continuity claim. Channel selects 0–7. Primary and Redundant cycle open, closed, short-simulated, unavailable. Apply timestamps and submits the selected channel while preserving the other seven entries.

6.1 Three dependent wins

The short opening journey makes each new idea depend on the preceding one:

1. On channel 0, submit Closed/Closed. One steady reward lamp confirms matching usable evidence.
2. On channel 1, submit Open/Closed. Two reward lamps confirm that disagreement is distinct from either usable state.
3. On channel 2, submit Unavailable/Closed. All eight lamps pulse for the final reward, confirming that missing evidence dominates disagreement.

Submitting a later answer early produces its ordinary state cadence but does not unlock a reward. Each reward lasts 1.5 seconds, then the panel returns to the channel assessments. The panel always shuts down two minutes after startup, independent of button input.

State	Indicator	Interpretation
open	off	matching open selections
closed	steady	matching closed selections
short-simulated	1 s on / 1 s off	matching synthetic short selection
stale	200 ms pulse each second	last accepted observation is too old
contradictory	200 ms alternating	usable observations disagree
unavailable	two 120 ms pulses per 2 s	one observation unavailable

7 Predict, observe, interpret

1. Predict the startup sweep, then power the panel and observe it.
2. Complete the three ordered wins above, predicting each reward before Apply.
3. After a reward clears, compare its channel cadence with the truth table and interpret only against the synthetic model.
4. Stop applying. Predict and observe stale after five seconds.
5. Leave the buttons untouched and observe every indicator release at the fixed two-minute ending.

Two separate checks. A successful startup response supports the claim that resources were acquired. It does not prove shutdown. The fixed ending releases every indicator; a dark LED alone is not electrical proof of high impedance.

8 Diagnosis

Observation	Likely cause	Next bounded check
no indicators ever respond	acquisition failure or wiring error	inspect D22–D29 and each resistor unpowered
button never changes selection	switch not reaching GND	observe TP28 high then low
unexpected contradictory cadence	selected values differ	cycle both banks to a known matching pair
all active channels become stale	no recent Apply event	apply one known pair and time five seconds
one LED disagrees with TP29	LED polarity or open resistor path	compare pin level, resistor, and diode orientation
shutdown LED dark but pin driven	light is not high-impedance proof	measure TP29 electrically

9 Go further

After the three wins, try Short-simulated/Short-simulated and wait for it to become stale. Then explain why a generic output callback would violate the inert boundary even if this particular sketch connected only LEDs. The repository’s automated tests cover rejected transactions, rollover, lifecycle, deterministic replay, compilation, and publication without turning those release checks into learner chores.

10 Reference trail

- [Searchable Lesson 028 HTML reference](#)
- [Public assessment and recorded-source interface](#)
- [Deterministic lifecycle, transaction, timing, and replay tests](#)
- [Lessons 028–030 safety and composition design](#)
- [Arduino Mega 2560 Rev3 datasheet](#)